

# PRAJWAL VENKAT VENKATESH

**Data Science | Credit Risk ML | Predictive Analytics | Statistical Modeling | Generative AI | MLOps**

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## SUMMARY

Data Scientist with 4+ years shipping production ML for credit risk, fraud detection, and Generative AI in regulated financial environments. Built a LangChain RAG pipeline now used by analysts, delivering a 3x speedup in research turnaround. Own end-to-end model development and MLOps with XGBoost, PySpark, Databricks, and MLflow, including model governance and compliance. Partner cross-functionally with business and technical stakeholders to translate model insights into decisions that move product, operations, and the bottom line.

## SKILLS

**Programming:** Python, SQL, C++, PySpark.

**Machine Learning & Statistics:** Predictive Modeling, Classification, Clustering, Feature Engineering, Supervised & Unsupervised Learning, XGBoost, LightGBM, CNN, Deep Learning, Anomaly Detection, A/B Testing, Hypothesis Testing.

**Frameworks & Libraries:** Pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex.

**Financial Analytics:** Risk Modeling, Credit Risk Analytics, Fraud Detection, Portfolio Analytics, Revenue Forecasting, Scenario Analysis, P&L Analysis, Stress Testing, Monte Carlo Simulation.

**Generative AI & NLP:** RAG Pipelines, LLMs, Prompt Engineering, Semantic Search, Vector Search, Natural Language Processing (NLP).

**Data Engineering & MLOps:** Apache Spark, Airflow, Kafka, ETL/ELT, Data Warehousing, Data Lake, MLflow, Docker, CI/CD, AWS, Databricks.

**Databases & Visualization:** Snowflake, PostgreSQL, BigQuery, MySQL, Power BI, Tableau.

**Risk & Governance:** Model Risk Management, Model Validation, Regulatory Reporting, Financial Controls.

**Other Skills:** Statistical Analysis, Time Series Forecasting, API Integration, Pipeline Orchestration, Agile Methodology.

## PROFESSIONAL EXPERIENCE

### JPMorgan Chase, United States | *Data Scientist, AI/ML*

**January 2025 – Present**

- Redesigned credit risk feature engineering workflows and retrained XGBoost models in Python against 95M+ lending records, improving portfolio default prediction AUC from 0.79 to 0.84 across retail and commercial portfolios.
- Implemented a RAG platform indexing SEC filings, earnings reports, and market disclosures securely stored in Amazon S3 using vector search and document embeddings, cutting analyst research time from 2.5 hours to under 50 minutes per request.
- Formulated comprehensive Monte Carlo simulations and scenario modeling frameworks using Python to replace legacy stress-testing processes, empowering risk teams to evaluate capital exposure across \$2B+ in monitored assets.
- Recalibrated anomaly detection models by leveraging MLflow monitoring to identify model drift, reducing false positive transaction screening alerts from 18% to 11% across enterprise payment processing channels.
- Automated predictive forecasting model deployment procedures across AWS cloud environments using Databricks and CI/CD pipelines, significantly strengthening analytical validation controls while reducing production rollout cycles from 3 days to 4 hours.
- Consolidated KPI reporting across Finance, Risk, and Portfolio Management stakeholders into a centralized Power BI dashboard covering 120+ performance indicators and directly informing quarterly investment planning decisions.

### Mphasis, India | *Data Scientist*

**June 2020 – July 2023**

- Built revenue forecasting solutions combining regression analysis and time series forecasting in Python, improving forecast accuracy from 74% to 89% across financial consulting engagements supporting strategic planning.
- Migrated fragmented batch ETL workflows to the Apache Spark framework, reducing processing runtime for 40M+ monthly records from 11 hours to under 3 hours across overarching enterprise warehouse ecosystems.
- Developed customer segmentation clustering models using behavioral and transactional attributes, identifying engagement gaps that increased qualified lead conversion rates from 12% to 19% across targeted campaigns.
- Created intelligent financial document processing workflows integrating LangChain, LlamaIndex, and GCP Vertex AI semantic retrieval architectures, decreasing regulatory compliance research effort by over 340 analyst hours annually.
- Introduced automated machine learning deployment standards using GitHub Actions and MLflow model monitoring, improving production release success rates from 82% to 97% across scalable cloud analytics environments.
- Engineered automated reporting infrastructure utilizing Tableau and SQL, consolidating P&L, portfolio, and regulatory metrics for 60+ stakeholders to reduce reporting latency from weekly to near-real-time and eliminate 8 hours of weekly manual preparation.

## EDUCATION

**Master of Science in Data Science (GPA: 4.0/4.0), The University of Texas at Arlington, Arlington, TX**

**May 2025**

**Coursework:** Data Science, Machine Learning, Big Data and Cloud Computing, Artificial Intelligence, Project Management.

## PROJECTS

### Quantitative Multi-Asset Portfolio Optimization & Rebalancing Engine

- Engineered a quantitative portfolio optimization framework in Python and PyTorch, querying 6M+ tick-level market records in Snowflake to compute covariance matrices and boost Sharpe ratio from 1.15 to 1.82 across multi-asset portfolios.
- Orchestrated containerized data ingestion pipelines using Apache Airflow and Docker to process high-frequency market feeds, slashing portfolio rebalancing calculation latency from 45 seconds to under 1.8 seconds across 500+ asset tickers.